WORKSHEET

Chapter 4: First order recurrence relations

1 Find the first five terms of the sequences defined by each of the following recurrence relations.

a $t_{n+1} = t_n - 3, t_1 = 10$

b $t_{n+1} = 5t_n, t_1 = 2$

c $t_{n+1} = 20 - t_n, t_1 = 15$

d $P_{n+1} = 2P_n + 1, P_1 = 4$

e $w_{n+1} = 2 - 2w_n, w_1 = 2$

f $u_{n+1} = -3u_n, u_1 = 5$

2 **a** If $u_{n+1} = u_n - 4$ and $u_2 = 9$, find u_4.

b If $t_{n+1} = t_n + 2.5$ and $t_3 = 8.5$, find t_5.

c If $t_{n+1} = 2t_n + 1$ and $t_5 = 31$, find t_7.

d If $u_{n+1} = 3 - 2u_n$ and $u_4 = -7$, find u_6.

e If $P_{n+1} = -2P_n$ and $P_2 = -6$, find P_4.

f If $w_{n+1} = 4w_n + 1$ and $w_3 = -3$, find w_5.

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3 **a** If $u_{n+1} = u_n + 7$ and $u_3 = 15$, find u_1.

b If $t_{n+1} = 10 - t_n$ and $t_3 = 6$, find t_1.

c If $u_{n+1} = 2u_n - 4$ and $u_5 = 8$, find u_3.

d If $w_{n+1} = 3w_n + 3$ and $w_4 = 66$, find w_2.

e If $t_{n+1} = \frac{1}{2}t_n + 4$ and $t_4 = 27$, find t_2.

f If $k_{n+1} = 4k_n + 3$ and $k_7 = -1$, find k_5.

4 Write the recurrence relation that defines the sequence:

a 3, 7, 11, 15, …

b 35, 28, 21, 14, …

c −10, −4, 2, 8, …

d 100, 50, 25, 12.5, …

e 60, 45, 30, 15, …

f 5, −6, −17, −28, …

5 The sequence 2, 7, 17,… is generated from the recurrence relation:

$t_{n+1} = rt_n + 3$ with $t_1 = 2$.

a Find the value of r.

b Find the fifth term in the sequence.

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6 The sequence 5, 11, 29, … is generated from the recurrence relation:

$t_{n+1} = rt_n - 4$ with $t_1 = 5$.

a Find the value of r.

b Find the fifth term in the sequence.

7 The sequence 10, −4, 10, … is generated from the recurrence relation:

$w_{n+1} = -1w_n + d$ with $w_1 = 10$.

a Find the value of d.

b Find the sixth term in the sequence.

8 The sequence 0, 4, 16, … is generated from the recurrence relation:

$u_{n+1} = 3u_n + d$ with $u_1 = 0$.

a Find the value of d.

b Find the fifth term in the sequence.

9 The first three terms of the sequence defined by the recurrence relation:

$t_{n+1} = rt_n + d$ with $t_1 = a$

are 3, 6 and 18.

Find the values of r, d and a.

10 The first three terms of the sequence defined by the recurrence relation:

$t_{n+1} = rt_n + d$ with $t_1 = a$

are 6, 17, and 39.

Find the values of r, d and a.

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11 The nth term of a sequence is given by the rule $t_n = 2^{n-1} + 3$.

a Find the first three terms in the sequence.

b Find the recurrence relation that defines the sequence.

12 The nth term of a sequence is given by the rule $t_n = 3^n + 1$.

a Find the first three terms in the sequence.

b Find the recurrence relation that defines the sequence.

13 The nth term of a sequence is given by the rule $t_n = 2^{n+1} - 7$.

a Find the first three terms in the sequence.

b Find the recurrence relation that defines the sequence.

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Answers

1 **a** 10, 7, 4, 1, −2

b 2, 10, 50, 250, 1250

c 15, 5, 15, 5, 15

d 4, 9, 19, 39, 79

e 2, −2, 6, −10, 22

f 5, −15, 45, −135, 405

2 **a** 1

b 13.5

c 127

d −31

e −24

f −43

3 **a** 1

b 6

c 5

d 6

e 84

f −1

4 **a** $t_{n+1} = t_n + 4, t_1 = 3$

b $t_{n+1} = t_n - 7, t_1 = 35$

c $t_{n+1} = t_n + 6, t_1 = -10$

d $t_{n+1} = \frac{1}{2}t_n, t_1 = 100$

e $t_{n+1} = t_n - 15, t_1 = 60$

f $t_{n+1} = t_n - 11, t_1 = 5$

5 **a** 2

b 77

6 **a** 3

b 245

7 **a** 6

b −4

8 **a** 4

b 160

9 $r = 4, d = -6, a = 3$

10 $r = 2, d = 5, a = 6$

11 **a** 4, 5, 7

b $t_{n+1} = 2t_n - 3, t_1 = 4$

12 **a** 4, 10, 28

b $t_{n+1} = 3t_n - 2, t_1 = 4$

13 **a** −3, 1, 9

b $t_{n+1} = 2t_n + 7, t_1 = -3$

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